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Wireless device activation tracking of a wireless satellite service

Abstract

Apparatuses, methods, and systems for operating a wireless satellite service are disclosed. One method includes maintaining, by a satellite service server, a database of an activation status of each of a plurality of subscriber devices as provided by a plurality of terrestrial service providers, receiving, by a satellite network, a request from a wireless device to access the wireless satellite service through a satellite link, wherein the satellite network is electronically connected to the satellite service server, querying, by the satellite network, the satellite service server to confirm whether the wireless device requesting access to the wireless satellite service is a subscriber device that is activated, providing, by the satellite service server, the activation status of the wireless device to the satellite network, and performing, by the satellite network, a wireless device attachment process with a one of the plurality terrestrial service providers based upon the activation status.

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Background/Summary

FIELD OF THE DESCRIBED EMBODIMENTS

(1) The described embodiments relate generally to wireless communications. More particularly, the described embodiments relate to systems, methods, and apparatuses for wireless device activation tracking of a wireless satellite service.

BACKGROUND

(2) Satellite networks operate to cover large areas and can have a large number of users. Further, only limited amounts of spectral resources are allocated to satellite networks. Accordingly, satellite networks should be selective in allowing user access to the satellite networks.

(3) It is desirable to have methods, apparatuses, and systems for wireless device activation tracking of a wireless satellite service.

SUMMARY

(4) An embodiment includes a method of operating a wireless satellite service. The method includes maintaining, by a satellite service server, a database of an activation status of each of a

plurality of subscriber devices as provided by a plurality of terrestrial service providers, receiving, by a satellite network, a request from a wireless device to access the wireless satellite service through a satellite link, wherein the satellite network is electronically connected to the satellite service server, querying, by the satellite network, the satellite service server to confirm whether the wireless device requesting access to the wireless satellite service is a subscriber device that is activated, providing, by the satellite service server, the activation status of the wireless device to the satellite network, and performing, by the satellite network, a wireless device attachment process with a one of the plurality terrestrial service providers based upon the activation status.

(5) Another embodiment includes a satellite system. The system includes a satellite network, and a satellite service server electronically connected to the satellite network. The satellite service server is configured to maintain a database of an activation status of each of a plurality of subscriber devices as provided by a plurality of terrestrial service providers. The satellite network is configured to receive a request from a wireless device to access the wireless satellite service through a satellite link, wherein the satellite network is electronically connected to the satellite service server and query the satellite service server to confirm whether the wireless device requesting access to the wireless satellite service is a subscriber device that is activated. The satellite service server is further configured to provide the activation status of the wireless device to the satellite network, and the satellite network is further configured to perform a wireless device attachment process with a one of the plurality terrestrial service providers based upon the activation status.

(6) Other aspects and advantages of the described embodiments will become apparent from the following detailed description, taken in conjunction with the accompanying drawings, illustrating by way of example the principles of the described embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows a satellite network interfaced with a plurality of terrestrial service providers, according to an embodiment.
- (2) FIG. 2 shows a satellite network in which a wireless device successfully attaches to the satellite network, and when the wireless device is not able to attach to the satellite network, according to an embodiment.
- (3) FIG. 3 shows a time-line of events in which a wireless device successfully attaches to the satellite network, according to an embodiment.
- (4) FIG. 4 shows a time-line of events in which an attachment request of a wireless device to the satellite network is rejected, according to an embodiment.
- (5) FIG. 5 shows a time-line of events of a passing early attachment check by a base station of a satellite network, according to an embodiment.
- (6) FIG. 6 shows a time-line of events of a failing early attachment check by a base station of a satellite network, according to an embodiment.
- (7) FIG. 7 shows a time-line of events of two different cases of maintaining activation statuses of subscriber devices, according to an embodiment.
- (8) FIG. 8 shows a time-line of events of a satellite service server tracking activation attempts by a wireless device, and further, requesting to block the wireless device after a set number of activation check failures, according to an embodiment.
- (9) FIG. 9 is a table that includes exemplary entries in a database of the satellite service server, according to an embodiment.
- (10) FIG. 10 is a flow chart that includes steps of a method for wireless device activation tracking of a wireless satellite service.

DETAILED DESCRIPTION

(11) The embodiments described include methods, apparatuses, and systems for wireless device activation tracking of a wireless satellite service.

(12) FIG. 1 shows a satellite network **140** interfaced with a plurality of terrestrial service providers **110, 120**, according to an embodiment. For an embodiment, a satellite service server **150** is configured to maintain a database **160** of an activation status of each of a plurality of subscriber devices as provided by the plurality of terrestrial service providers **110, 120**. The terrestrial service providers **110, 120** maintain databases **125, 126** that include customer information.

(13) For an embodiment, the satellite network **140** is configured to receive a request from a wireless device **130** to access the wireless satellite service **150** through a satellite link **172** supported by a satellite **170**. As shown, the satellite network **140** is electronically connected to the satellite service server **150**. Further, the satellite network **140** and the wireless satellite service **150** are electronically connected to the plurality of terrestrial service providers **110, 120**.

(14) For an embodiment, the satellite network **140** is further configured to query the satellite service server **150** to confirm whether the wireless device **130** requesting access to the wireless satellite service is a subscriber device that is activated per the database **160**.

(15) For an embodiment, the satellite service server **150** is configured to provide the activation status of the wireless device **150** to the satellite network **140** in response to the query. For an embodiment, the activation status of the wireless device **130** is active or non-active.

(16) For an embodiment, the satellite network **140** is further configured to perform a wireless device **130** attachment process when the activation status indicates the wireless device **130** is determined to be active. For an embodiment, this includes the satellite network **140** performing the wireless device **130** attachment process with a one of the plurality terrestrial service providers **110, 120** based upon the active status.

(17) It is to be understood that the satellite link **172** is a scarce resource. That is, there is only a limited amount of frequency spectrum allocated for use by the satellite link **172**. Accordingly, it can be desirable to only let devices of paid customers (subscribers) to access the wireless satellite service. For an embodiment, customers may purchase the subscription of the wireless satellite service as insurance in case needed but may not be needed. For an embodiment, wireless devices of non-prepaid users (non-customers) are not allowed to have access to the wireless satellite service.

(18) For an embodiment, the wireless device **130** attachment process refers to satellite network **140** attachment which includes authentication, security, device capability, and configuration message exchange between the wireless device **130** and the satellite network **140**. For an embodiment, when the wireless device roams to the satellite network **140**, the satellite network **140** authenticates with at least one of the terrestrial networks **110, 120**.

(19) For an embodiment, a subscriber device activation process includes checking the device's billing, registration and certification status (can include more checks) of the subscriber device and determining whether the subscriber device is allowed to access the network or not.

(20) For an embodiment, each terrestrial network **110, 120** includes a terrestrial core network **112, 122**, and a terrestrial activation/deactivation server **114, 124**. For an embodiment, the terrestrial core networks **112, 122** each routes traffic within the terrestrial network and controls access to the terrestrial network. For an embodiment, the terrestrial activation/deactivation servers **114, 124** each maintains the activation status of devices belonging to the terrestrial network.

(21) For an embodiment, the satellite network **140** includes a base station and a core network. For an embodiment, the base station is configured to receive the request from the wireless device **130** to access the wireless satellite service **150** through the satellite link **172**. For an embodiment, the core network is configured to route traffic and control access of wireless devices to the satellite network.

(22) For an embodiment, the satellite network **140** is operated by Company A which maintains the satellite service server **150**. For an embodiment, a terrestrial network (such as, terrestrial networks

110, 120) is operated by Company B which maintains a database (**125, 126**) of devices activated/unactivated on a service of company A. Company A also gets the activated/unactivated devices information and maintains it in its own database **160**. For an embodiment, company A and company B ensure that the database **160** of Company A has the most recent information using information push and pull mechanisms (that is, the satellite service server **150** of company A pulls the information or the terrestrial network or company B pushes the information to the satellite service server **150**). When a wireless device **130** tries to connect to the satellite network of company A, the satellite network **140** of company A first checks in its own database **160** before authenticating with the terrestrial network of company B. If the wireless device **130** is not activated based on the database **160** company A, then the satellite network **140** of company A doesn't go for a full authentication process (wherein full authentication process includes connecting with the terrestrial network of company B, checking the activation status from the database of the terrestrial network of company B, verifying and exchanging the security keys) with the terrestrial network of company B. Instead, the satellite network **140** of company A can directly reject the request to connect by the wireless device **130**. The described process here is valuable for satellite network, as the satellite network has limited bandwidth and the process described here helps in reducing the number of messages transmitted through satellite network, and therefore improves satellite network capacity. In other words, the embodiments described improve the satellite network by reducing the network attachment process overhead and also reduces latency.

(23) Existing networks perform the full authentication process with the terrestrial network of company B for every wireless device that tries to connect to the satellite network without checking their own database (since they don't maintain their own database). This increases overhead on the satellite network as the wireless devices which are not allowed (to use company A's service) also go through the full authentication process with the terrestrial network of company B via satellite network which is a scarce resource.

(24) For at least some of the described embodiments, the terrestrial network of company A provides appropriate rejection codes to the wireless devices and forces the wireless devices to reduce future connection attempt, and thereby further save resources (time and frequency spectrum) of the satellite network.

(25) FIG. 2 shows a satellite network in which a wireless device **230** successfully attaches (roaming network **202**) to the satellite network, and when the wireless device is not able to attach (roaming network **204**) to the satellite network, according to an embodiment. A first sequence of actions of the roaming network **202** of a wireless device **230** roaming and connecting to the satellite network **240** includes the wireless device **230** requesting access (attach request) **281** to the satellite network **240**. The first sequence of actions **202** further includes the satellite network **240** checking the activation status **282** which includes querying the satellite service server **250** to confirm whether the wireless device **230** requesting access to the wireless satellite service is a subscriber device that is activate as stored in a database **260** of the satellite network **240**. If the activation status query is a success (wireless device **230** is a subscriber device), the activation check results in a status active **283** being sent back to the satellite network **240**. Upon receiving the successful status active, the satellite network **140** is further configured to perform a wireless device **130** attachment success process **284**. That is, the satellite network **140** performs the wireless device **230** attachment process with the one of the plurality terrestrial service providers based upon the active status.

(26) For an embodiment, at least one terrestrial network **210** includes a server **214** that maintains the activation statuses of the subscriber devices. The activation statuses of the subscriber devices are stored in the database **260** of the satellite service server **250** and provided **285** to the satellite service server **250**. Further, a terrestrial core network **212** of the terrestrial network **210** maintains and provides **286** authentication information to the satellite network **240**.

(27) A second sequence of actions (roaming network **204**) of a wireless device **230** attempting to

roam and connect to the satellite network **240** of the satellite network **240** includes a wireless device **230** requesting access (attach request) **291** to the satellite network **240**. The second sequence of actions further includes the satellite network **240** checking the activation status **292** which includes querying the satellite service server **250** to confirm whether the wireless device **230** requesting access to the wireless satellite service is a subscriber device that is activated. If the activation status query is a failure (subscriber device not activated), the activation check results in a status not active **293** being sent back to the satellite network **240**. Upon receiving the status not active **293**, the satellite network **140** is further configured to fail the attachment success process **294**. For an embodiment, the authentication **286** with the terrestrial network **210** does not occur when receiving the status not active **293**.

(28) As shown, the terrestrial activation/deactivation server **214** of the terrestrial network **210** provides the activation status **285** to the satellite service server **250** which is stored in the database **260**.

(29) FIG. 3 shows a time-line of events in which a wireless device successfully attaches to the satellite network, according to an embodiment. A first event **301** includes a wireless device **330** completing an RRC (radio resource control) setup with a base station of a satellite network **310**. A second event **302** includes the wireless device **330** sending an attachment request to the base station of the satellite network **310**. A third event **303** includes the satellite network **310** performing an activation state check of the wireless device **330**. That is, the satellite network **310** queries the satellite service server **350** to determine whether the wireless device **330** is an activated subscriber of the terrestrial service provider **360**. If the wireless device **330** is determined to be an activated subscriber, then a fourth event **304** includes the satellite network **310** receiving the wireless device active status from the satellite service server **350**. A fifth event **305** includes the satellite network **310** attaching with the terrestrial network **360**. A sixth step **306** includes the satellite network **310** completing the attachment process with the wireless device **330**.

(30) For at least some embodiments, RRC connection setup includes a connection establishment process between base station of the satellite network **310** and the wireless device **330**. A typical RRC connection establishment includes a prach (periodic random access channel also referred to as message 1) transmission by the wireless device **330**. The base station responds with RAR (random access response, also referred to as message 2). The wireless device **330** then sends (transmits) an RRC connection request message (referred to as message 3), and the base station sends (transmits) an RRC connection setup message (also referred to as message 4) to the wireless device **330**. The wireless device then responds with an RRC connection complete message (message 5) to base station. As will be described, for some embodiments, the RRC connection setup message (also referred to as message 4) and the RRC connection complete message (message 5) may be eliminated when the wireless device is determined not to be activated.

(31) FIG. 4 shows a time-line of events in which an attachment request of a wireless device **430** to the satellite network **410** is rejected, according to an embodiment. A first event **401** includes a wireless device **430** completing an RRC (radio resource control) setup with a base station of a satellite network **410**. A second event **402** includes the wireless device **430** sending an attachment request to the base station of the satellite network **410**. A third event **403** includes the satellite network **410** performing an activation state check of the wireless device **430**. That is, the satellite network **410** queries the satellite service server **450** to determine whether the wireless device **430** is an activated subscriber of the terrestrial service provider **460**. If the wireless device **430** is determined to not be an active subscriber, then a fourth event **404** includes the satellite network **410** receiving the wireless device not-activated status from the satellite service server **450**. Note that in this situation, the satellite network **410** does not perform attachment with the terrestrial network **460**. A fifth step **405** includes the satellite network **410** sending an attachment rejection **405** to the wireless device **406** along with a rejection code.

(32) FIG. 5 shows a time-line of events of a passing early attachment check by a base station of a

satellite network, according to an embodiment. This time-line includes an RRC setup process **570** and a wireless device attachment process **580**. The embodiments of FIG. 5 and FIG. 6 include an early attachment check which reduces the impact on satellite resources during the wireless device connection (attachment) process. Specifically, through the novel RRC attachment process, determination is made earlier whether an attaching wireless device has been activated or not. If a wireless device is determined not to be activated, then several messages (**574**, **575**, **302**, **306**) can be eliminated from being transmitted over the wireless satellite link between the wireless device and the base station of the satellite network, thereby reducing the impact on the valuable and scarce satellite connection resources.

(33) A first event **571** includes the satellite network **510** receiving a PRACH (periodic random-access channel) request from the wireless device **530**. A second event **572** includes the satellite network **510** replying with a RAR (random access response). A third step **573** includes the satellite network **510** receiving an RRC request (also referred to as “message 3”) from the wireless device **530**. Upon receiving the RRC request of step **573**, the device activation check of step **303** is performed. If the wireless device is determined to be activated, the device active step **304** is received by the core of the satellite network. That is, upon receiving the RRC connection request of step **573** from the wireless device **530**, the satellite network **51** sends a device activation check (query) **303** to the satellite server **550**. If the wireless device **530** is determined to be a subscriber device, the satellite server **550** responds with an indicator (device active **304**) that the wireless device is an active device.

(34) A fourth step **574** includes the satellite network **510** sending an RRC setup back to the wireless device **530** after receiving the device active indication of step **304**. A fifth step **575** includes the satellite network **510** receiving an RRC connection complete from the wireless device **530**.

(35) The wireless device then sends the attach request **302**. The satellite network **510** then attaches with the terrestrial network **360**. Finally, the satellite network **510** completes the attachment process **306** with the wireless device **530**.

(36) FIG. 6 shows a time-line of events of a failing early attachment check by a base station of a satellite network **610**, according to an embodiment. Here, the RRC setup process **570** is the same as described for FIG. 5 except steps **574**, **575** are not performed or executed if the wireless device is determined not to be active through step **404**. Here, the attachment process includes the wireless device **630** being determined not to be an active device in response to the activation check **303** sent by the satellite network **610** to the satellite service server **650**. The device not active status **404** is communicated to the satellite network **610**. The satellite network **610** then communicates an RRC connection rejection **606** to the wireless device **630**. Note that in this situation, the satellite network **610** does not perform attaching with the terrestrial network **660**, and steps **302**, **306**, **574**, **575** are not performed. The elimination of these steps is novel and saves satellite network resources (time and frequency allocations) of the wireless satellite link between the wireless device **330** and the base station of the satellite network **310**.

(37) As previously described, the embodiments of FIG. 5 and FIG. 6 include embodiments of an early attachment check which reduces the impact on satellite resources during the wireless device connection (attachment) process. Specifically, through the novel RRC attachment process, determination is made earlier whether an attaching wireless device has been activated or not. After step **573** that includes the RRC request (message 3), activation of the wireless device is checked through step **303**. If the wireless device is determined not to be activated (step **404**), then several messages (of steps **574**, **575**, **302**, **306**) can be eliminated from being transmitted over the wireless satellite link between the wireless device and the base station of the satellite network, thereby reducing the impact on the valuable and scarce satellite connection resources.

(38) At least some embodiments further include sending, by the satellite network, a rejection code to the wireless device based upon the activation status of the wireless device in the satellite service server, wherein a retry procedure of the wireless device is based upon the rejection code. For an

embodiment, the rejection code specifies a period of time (for example, 24 hours) during which the wireless device cannot retry another connection attempt. For an embodiment, if the wireless device has been determined to be stolen, damaged, or not operating within bounds the rejection code may permanently disable the wireless device. This may be done by deleting information from the SIM card such as PLMN (Public Land Mobile Network) ID.

(39) FIG. 7 shows a time-line of events of two different cases **701**, **702** of maintaining activation statuses of subscriber devices, according to an embodiment. For the first case **701**, the satellite service server **750** submits a request for the activation status of a connecting wireless device. The terrestrial service provider **760** responds with the activation status which is then stored in the database of the terrestrial service provider **760**.

(40) For the second case, the terrestrial service provider **760** continually provides the activation/deactivation status of each of the subscriber devices over time to the satellite service server **750**.

(41) FIG. 8 shows a time-line of events of a satellite service server **850** tracking activation attempts by a wireless device, and further, requesting to block the wireless device after a set number of activation check failures, according to an embodiment. For at least some embodiments, this enables the satellite network (through the satellite service server **850**) to notify the terrestrial network (through the terrestrial service provider **860**) that a wireless device is making excessive demands on the satellite network by repeatedly requesting connection despite being inactive or otherwise not eligible to use the satellite network. Process **801** shows the satellite service server **850** providing the terrestrial service provider **860** with notification of connection and attachment requests by the wireless device. Process **802** shows the terrestrial service provider **860** providing a request or instructions to the satellite service provider **850** to block the wireless device for future satellite network connections. The block for be for any number of future attempts by the wireless device or for any selected amount of time.

(42) FIG. 9 is a table that includes exemplary entries in a database **260** of the satellite service server, according to an embodiment. Exemplary entries to the database **260** include an IMSI (international mobile subscriber identity) of the customer device, IMEI. activation status and Quota/Plan definition, 'Time when device was activated', last terrestrial connection time. last satellite connection time, location of the device, device's temporary identifier, device RRC connection and attach and security context, satellite network usage statistics (#of msgs, msg size, Frequency, UL/DL data transfer statistics). terrestrial to satellite roaming events, connection attempts and frequency while in deactivated state. The maintenance of such a database (table) external to the core of the satellite network is unique to existing satellite networks.

(43) FIG. 10 is a flow chart that includes steps of a method for wireless device activation tracking of a wireless satellite service. A first step **1010** includes maintaining, by a satellite service server, a database of an activation status of each of a plurality of subscriber devices as provided by a plurality of terrestrial service providers. A second step **1020** receiving, by a satellite network, a request from a wireless device to access the wireless satellite service through a satellite link, wherein the satellite network is electronically connected to the satellite service server. A third step **1030** querying, by the satellite network, the satellite service server to confirm whether the wireless device requesting access to the wireless satellite service is a subscriber device that is activated. A fourth step **1040** includes providing, by the satellite service server, the activation status of the wireless device to the satellite network. A first step **1050** includes performing, by the satellite network, a wireless device attachment process with a one of the plurality terrestrial service providers based upon the activation status.

(44) As previously described, satellite wireless links are a scarce resource. That is, there is only a limited amount of frequency spectrum allocated for use by the satellite links. Accordingly, it can be desirable to only let devices of paid customers (subscribers) to access the wireless satellite service. For an embodiment, customers may purchase the subscription of the wireless satellite service as

insurance in case needed but may not be needed. For an embodiment, wireless devices of non-prepaid users (non-customers) are not allowed to have access to the wireless satellite service.

(45) Further, as previously described, for an embodiment, the wireless device attachment process refers to satellite network attachment which includes authentication, security, device capability, and configuration message exchange between the wireless device **130** and the satellite network **140**. For an embodiment, when the wireless device roams to the satellite network **140**, the satellite network **140** may authenticate with at least one of the terrestrial networks **110**, **120**.

(46) For an embodiment, the activation process includes checking the device's billing, registration and certification status (can include more checks) of the wireless device and determining whether the wireless device is allowed to access the satellite network or not. As previously described, the satellite network includes a satellite base station and a satellite core network.

(47) At least some embodiments further include receiving, by the satellite service server, activation status change notifications from the plurality of terrestrial service providers indicating when a subscriber device subscription to the wireless satellite service is activated or deactivated. The database of the activation status of each of a plurality of subscriber devices as provided by a plurality of terrestrial service providers can be accordingly updated.

(48) At least some embodiments further include performing, by the satellite network, an authentication with at least one of the plurality of the terrestrial service providers for the wireless device during the attachment process as shown in FIG. 3.

(49) At least some embodiments further include rejecting, by the satellite network, the access request of the wireless device when the satellite service server returns a non-active status when queried by the satellite network as shown in FIG. 4. For an embodiment, this is a case in which there is a failure in response to the first authentication attempt. As previously described, at least some embodiments further include sending, by the satellite network, a rejection code to the wireless device based upon the activation status of the wireless device in the satellite service server, wherein a retry procedure of the wireless device is based upon the rejection code. For an embodiment, the rejection code specifies a period of time (for example, 24 hours) in which the wireless device cannot retry another connection attempt. For an embodiment, if the wireless device has been determined to be stolen, damaged, or not operating within bounds the rejection code may permanently disable the wireless device. This may be done by deleting information from the SIM card such as PLMN (Public Land Mobile Network) ID.

(50) At least some embodiments further include sending, by the wireless satellite service, a message to the wireless device providing information about the activation of the wireless device in the satellite service server. For an embodiment, the message includes information about a data quota of the wireless device and whether the data quota is about to expire. For an embodiment, the message includes information indicating that the wireless device is inactive and that only emergency services will work. For an embodiment, the message includes information indicating that the wireless device is roaming into a region where service cannot be provided. For example, the wireless device may have purchased only European coverage, and is now located in the United States. For an embodiment the message may be contained inside the network rejection code.

(51) At least some embodiments further include granting network access by the satellite network to the wireless device for on a limited basis based upon the rejection code. For an embodiment, the limited basis includes granting access for a limited time duration. For an embodiment, the limited basis includes granting access for a limited set of services/messages. For an embodiment, the limited basis includes granting access for a limited set of service locations. The described embodiment of the limited basis can have variations (such as, the variable time) that can be selected, for example, per policy of satellite network provider or legal requirements (for example, might be required to provide SOS).

(52) At least some embodiments further include sending, by a satellite core network of the satellite network, a message to the wireless device through a terrestrial wireless network based upon the

rejection code. Sending the message to the wireless device through the terrestrial network reduces messaging over the wireless satellite link and is generally more cost effective. Accordingly, the message can include more detail than a short terse message sent over the satellite wireless link. For at least some embodiments, the message includes further elaboration of why a rejection was given. Further, for an embodiment, the message may include a solicitation for service reactivation. Further, for an embodiment, the message may include a control command to the device to engage in a different mode of operation. These messages may be sent at both current and future times.

(53) At least some embodiments further include receiving, by the satellite network, identification information (such as, IMSI (international mobile subscriber identity)) from the wireless device. For an embodiment, the identification information is received by the satellite network from the wireless device during a radio resource control connection (RRC) process. For an embodiment, the RRC is a connection process a wireless device makes with a base station. For an embodiment, network attachment and authentication is a process the wireless device makes with a core network that comes after the RRC connection process has successfully concluded, and the attach process uses the RRC connection that was established.

(54) At least some embodiments further include sending, by the satellite network a rejection code to the wireless device based upon the activation status of the wireless device in the satellite service server and rejecting the radio resource connection of the wireless device based upon the activation status of the wireless device as stored in the database of the satellite service server.

(55) At least some embodiments further include authenticating, by the satellite network, with at least one of the plurality of terrestrial service providers based on an activation status of the wireless device maintained by the database. At least some embodiments further include authenticating based on the identification information (IMSI and other) provided by the wireless device during the authentication process. This is a success for the first authentication, and now doing 2nd authentication with the partner terrestrial network by the satellite network. As previously described, the opportunity to reject the request earlier in the process allows for preservation scarce satellite channel resource by not sending RRC setup (574 of FIG. 5), RRC connection complete (575 of FIG. 5), attach request (302 of FIG. 3), and attach rejection (406 of FIG. 4) messages.

(56) At least some embodiments further include allowing, by the satellite network, the wireless device to access a network of the wireless satellite service after receiving an authentication response from at least one of the plurality of terrestrial service providers. This includes a pass double authentication.

(57) At least some embodiments further include allowing, by the satellite network, the wireless device to access a network of the wireless satellite service after receiving an activation status response from the satellite service server. This includes a pass, null authentication case.

(58) For an embodiment, the wireless device requesting access to the wireless satellite service has a device ID (IMSI (international mobile subscriber identity)) that is not an ID of the wireless satellite service, thereby requiring the satellite service server to track and authenticate the wireless device. For an embodiment, some device IDs belong to the satellite service provider, and some belong (owned, issued by, etc.) to the plurality of terrestrial service providers.

(59) For an embodiment, the wireless device requesting access to the wireless satellite service has a device ID (IMSI) that is an ID of the wireless satellite service and is programmed into a SIM (Subscriber Identity Module) card of the wireless device.

(60) For an embodiment, the satellite service server further contains data quota information for roaming wireless devices, wherein roaming devices are wireless devices that roam from a terrestrial network to the satellite network. For an embodiment, the quota information includes a data quota received from the terrestrial network. For an embodiment, the satellite network operates to reject or accept requests from wireless devices based on the data quota. For an embodiment, a rejection may include a sleep reject based on a quota scheme. For at least some embodiments, the wireless satellite service normally does not know what service plan a wireless device has if it is provided by

a terrestrial network. However, the terrestrial network can provide some of this information to the satellite service server and the wireless satellite service can make more informed decisions based upon this information.

(61) For an embodiment, the satellite service server maintains a count of the number of access attempts from a wireless device. For an embodiment, a core network of the satellite network sends a rejection code to the wireless device through a terrestrial network based upon the authentication attempt count.

(62) For an embodiment, the satellite service server further contains security features for roaming wireless devices. For an embodiment, the security features are received from the terrestrial network. For an embodiment, the satellite network rejects or accepts an access request by a wireless device based on the security features.

(63) For an embodiment, the satellite service server further contains location information for roaming wireless devices. For an embodiment, the location information is received from the terrestrial network. For an embodiment, the satellite network rejects or accepts an access request by a wireless device based on the location information.

(64) An alternative embodiment includes the steps of the method for wireless device activation tracking of a wireless satellite service from the perspective of the satellite service server.

Specifically, for an embodiment, the method includes maintaining, by a satellite service server, a database of an activation status of each of a plurality of subscriber devices as provided by a plurality of terrestrial service providers, wherein a satellite network receives a request from a wireless device to access the wireless satellite service through a satellite link, wherein the satellite network is electronically connected to the satellite service server. Further, the method includes receiving, by the satellite service server, a querying from the satellite network to confirm whether the wireless device requesting access to the wireless satellite service is a subscriber device that is activated, and providing, by the satellite service server, the activation status of the wireless device to the satellite network, wherein the satellite network performs a wireless device attachment process with a one of the plurality terrestrial service providers based upon the activation status.

(65) Although specific embodiments have been described and illustrated, the embodiments are not to be limited to the specific forms or arrangements of parts so described and illustrated. The described embodiments are to only be limited by the claims.

Claims

1. A method of operating a wireless satellite service, comprising: maintaining, by a satellite service server, a database of an activation status of each of a plurality of subscriber devices as provided by a plurality of terrestrial service providers; receiving, by a satellite network, a request from a wireless device to access the wireless satellite service through a satellite link, wherein the satellite network is electronically connected to the satellite service server, comprising receiving, by the satellite network, a PRACH (periodic random-access channel) request from the wireless device; replying, by the satellite network, with a RAR (random access response); receiving, by the satellite network, an RRC (radio resource control) request from the wireless device; querying, by the satellite network, the satellite service server to confirm whether the wireless device requesting access to the wireless satellite service is a subscriber device that is activated upon receiving the RRC request, comprising: sending, by the satellite network, a device activation check (query) to the satellite server; providing, by the satellite service server, the activation status of the wireless device to the satellite network; responding, by the satellite network, with an indicator that the wireless device is an active device when the wireless device is determined to be a subscriber device, comprising: a. sending, by the satellite network, an RRC setup back to the wireless device after receiving the device active indication; b. receiving, by the satellite network, an RCC connection complete from the wireless device; responding, by the satellite network, with a device not active

indicator when the wireless device is determined not to be a subscriber device, thereby bypassing steps a and b to reduce an impact on satellite connection resources when responding with the device not active indicator.

2. The method of claim 1, wherein the satellite network includes a satellite base station and a satellite core network.

3. The method of claim 1, further comprising: receiving, by the satellite service server, activation status change notifications from the plurality of terrestrial service providers indicating when a subscriber device subscription to the wireless satellite service is activated or deactivated.

4. The method of claim 1, further comprising performing, by the satellite network, an authentication with at least one of the plurality of the terrestrial service providers for the wireless device during an attachment process.

5. The method of claim 1, further comprising; rejecting, by the satellite network, the access request of the wireless device when the satellite service server returns a non-active status when queried by the satellite network.

6. The method of claim 5, further comprising: sending, by the satellite network, a rejection code to the wireless device based upon the activation status of the wireless device in the satellite service server, wherein a retry procedure of the wireless device is based upon the rejection code.

7. The method of claim 1, further comprising sending, by the wireless satellite service, a message to the wireless device providing information about the activation of the wireless device in the satellite service server.

8. The method of claim 6, further comprising granting network access by the satellite network to the wireless device for on a limited basis, wherein the rejection code specifies a period of time during which the wireless device cannot retry another connection attempt.

9. The method of claim 6, further comprising sending, by a satellite core network of the satellite network, a message to the wireless device through a terrestrial wireless network based upon the rejection code, wherein the message sent through the terrestrial network includes more detail than a message sent through the satellite network.

10. The method of claim 6, wherein the message sent through the terrestrial network includes a control command to the wireless device to engage in a different mode of operation.

11. The method of claim 10, further comprising: sending, by the satellite network a rejection code to the wireless device based upon the activation status of the wireless device in the satellite service server; rejecting the radio resource connection of the wireless device based upon the activation status of the wireless device as stored in the database of the satellite service server.

12. The method of claim 10, further comprising: authenticating, by the satellite network, with at least one of the plurality of terrestrial service providers based on an activation status of the wireless device maintained by the database.

13. The method of claim 4, further comprising: allowing, by the satellite network, the wireless device to access a network of the wireless satellite service after receiving an authentication response from the satellite service server.

14. The method of claim 1, further comprising: allowing, by the satellite network, the wireless device to access a network of the wireless satellite service after receiving an activation status response from the satellite service server.

15. The method of claim 1, wherein the wireless device requesting access to the wireless satellite service has a device ID (IMSI (international mobile subscriber identity)) that is not an ID of the wireless satellite service, thereby requiring the satellite service server to track and authenticate the wireless device.

16. The method of claim 1, wherein the wireless device requesting access to the wireless satellite service has a device ID (IMSI) that is an ID of the wireless satellite service and is programmed into a SIM (Subscriber Identity Module) card of the wireless device.

17. The method of claim 8 1, wherein the limited basis further includes granting access for a

limited set of service locations.

18. The method of claim 1, wherein the satellite service server maintains a count of the number of authentication attempts from a wireless device.

19. A satellite system, comprising: a satellite network; a satellite service server electronically connected to the satellite network; the satellite service server configured to: maintain a database of an activation status of each of a plurality of subscriber devices as provided by a plurality of terrestrial service providers; satellite network configured to: receive a request from a wireless device to access the wireless satellite service through a satellite link, wherein the satellite network is electronically connected to the satellite service server, comprising satellite network operating to: receive a PRACH (periodic random-access channel) request from the wireless device; reply with a RAR (random access response); receive an RRC (radio resource control) request from the wireless device; query the satellite service server to confirm whether the wireless device requesting access to the wireless satellite service is a subscriber device that is activated upon receiving the RRC request, comprising satellite network operating to: send a device activation check (query) to the satellite server; the satellite service server further configured to: provide the activation status of the wireless device to the satellite network; respond with an indicator that the wireless device is an active device when the wireless device is determined to be a subscriber device, comprising satellite network operating to: a. send an RRC setup back to the wireless device after receiving the device active indication; b. receive an RRC connection complete from the wireless device; respond with a device not active indicator when the wireless device is determined not to be a subscriber device, thereby bypassing steps a and b to reduce an impact on satellite connection resources when responding with the device not active indicator.

20. The satellite system of claim 19, wherein a core network of the satellite network sends the rejection code to the wireless device through a terrestrial network based upon the authentication attempt count.
